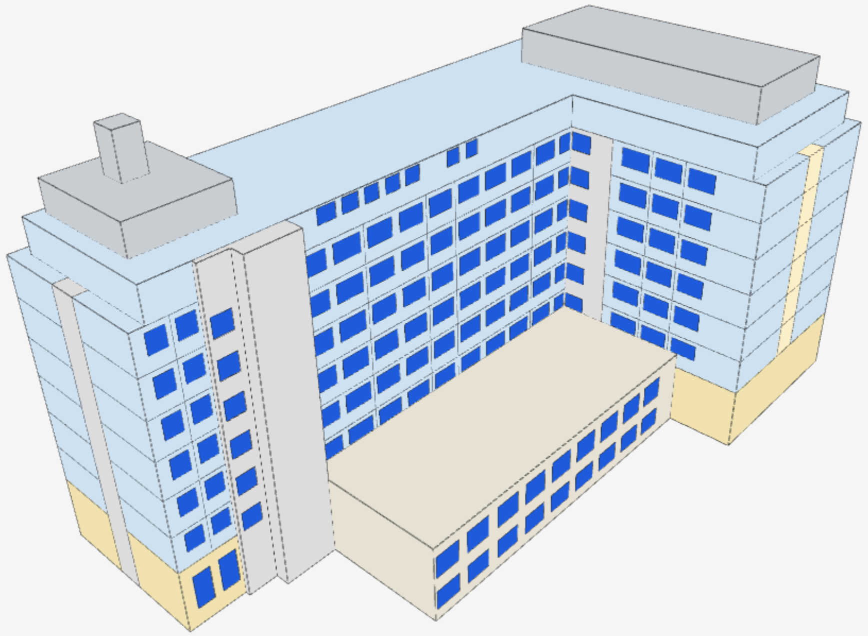
面向设计前期的

仿真轻量 BIM Agent

A lightweight BIM agent
for early-stage design simulation



多模态输入 → 轻量 BIM → 多物理场仿真

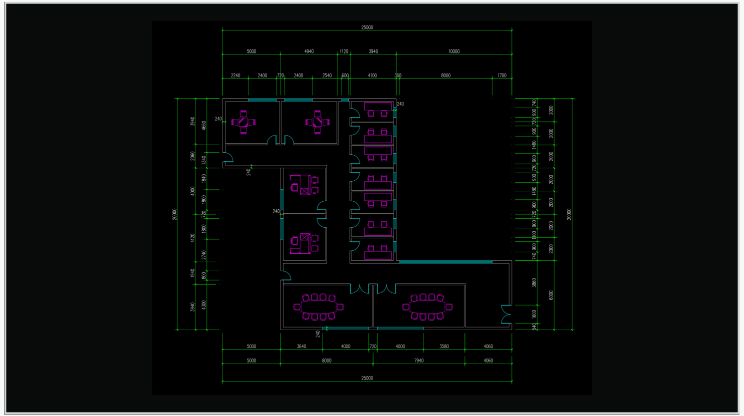
Multimodal inputs · Lightweight BIM · Multiphysics simulation

VOIMATALO

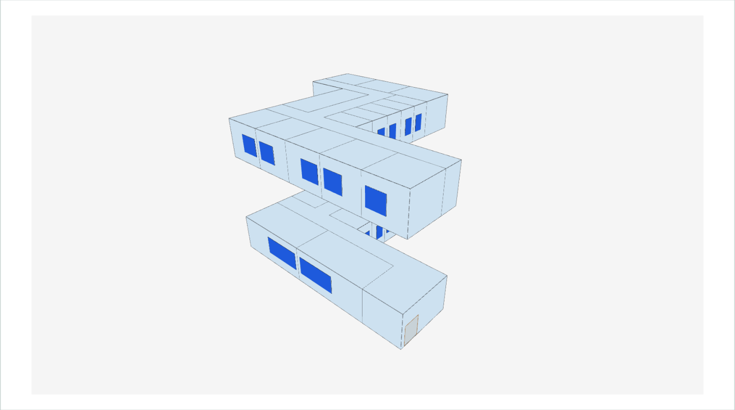
拖动旋转 / Drag to rotate

让建筑仿真跟上方案迭代

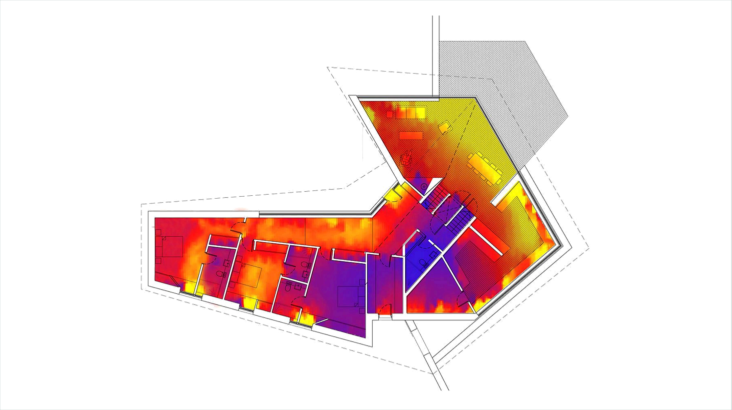
Bringing building simulation into the design iteration loop



人工识读
Interpret
→



专业配置
Configure
→



方案与图纸
Design & drawings

几何模型
Geometry model

仿真结果示例
Simulation result example

几何整理
Geometry cleanup
→
分区与物性
Zoning & properties
→
后端模型配置
Backend setup
→
计算与结果回传
Run & return results

←
长链路 · 依赖人工 · 反馈滞后
Long workflow · Manual effort · Delayed feedback

研究目标
Our aim

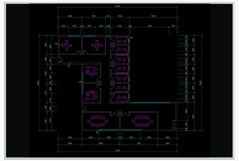
建立轻量的公共仿真底座

A shared lightweight foundation for building simulation

设计 ⇄ 轻量 BIM ⇄ 仿真
Design · Lightweight BIM · Simulation

多模态输入，轻量 BIM 输出

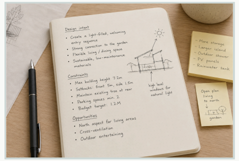
Multimodal inputs to lightweight BIM



图纸 / Drawings

图像 / Images

三维体量 / 3D meshes

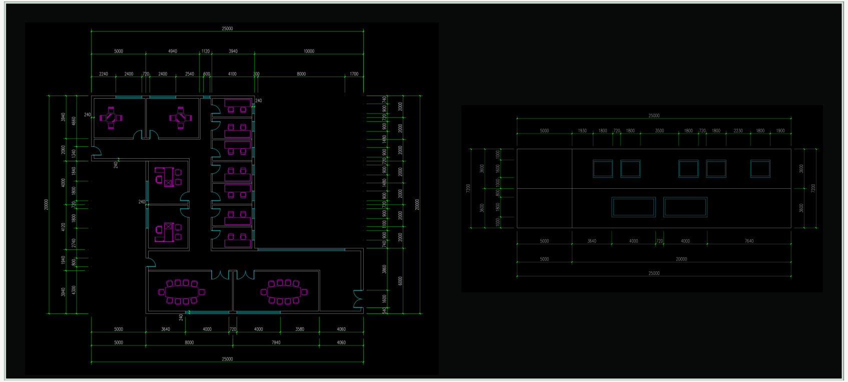


文字意图 / Design intent

↓

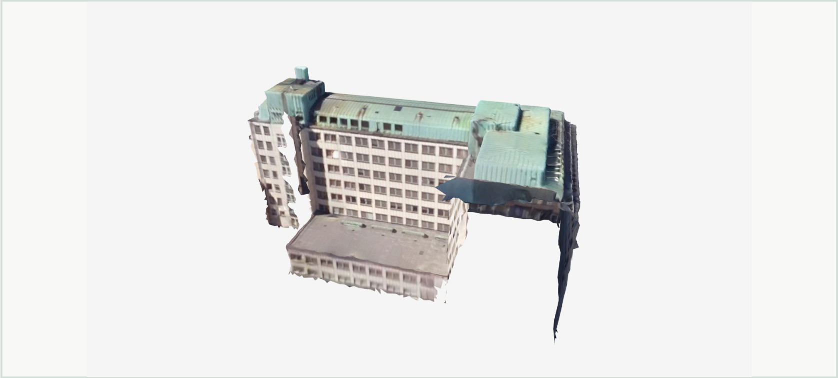
BIM Agent

多模态建模流程 / Multimodal modelling pipeline



还原建模

Reconstruction



部分推理建模

Partial inference

完全推理建模

Full inference

↓ 轻量 BIM Lightweight BIM

从信息理解，到公共仿真底座

From input understanding to a shared simulation foundation

BIM Agent

目标与输入 Goals & inputs

01
输入识读
Perceive inputs

→

02
信息整合
Align information

→

03
空间推理
Reason about spaces

→

04
几何生成
Build geometry

→

05
检查修正
Check & refine

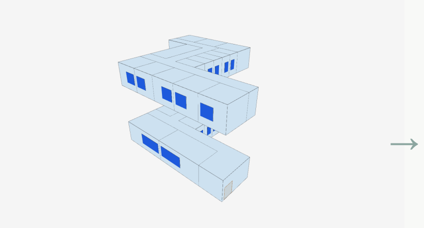
根据模型与工具反馈迭代 Iterate with model & tool feedback

模型能力
Model capabilities

几何工具
Geometry tools

交互补充
User input

→



轻量 BIM
Lightweight BIM

→

仿真插件 Simulation plugins

EnergyPlus
热与能耗 / Thermal & energy
● 已接通 / Connected

扩展接口 / Extensible interfaces

Radiance
采光 / Daylighting

OpenFOAM
气流 / Airflow

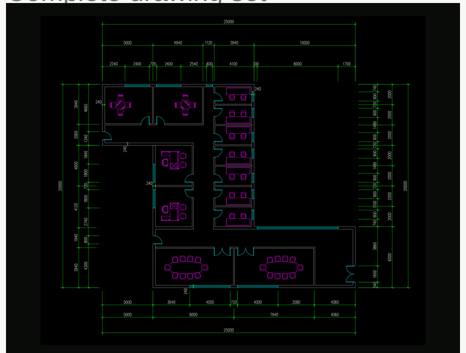
Pachyderm
声学 / Acoustics

完整建筑图纸 → 轻量 BIM

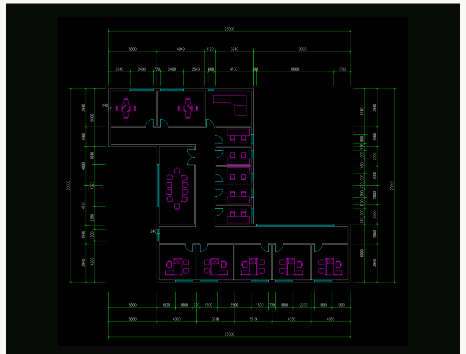
A complete architectural drawing set to lightweight BIM

完整图纸输入

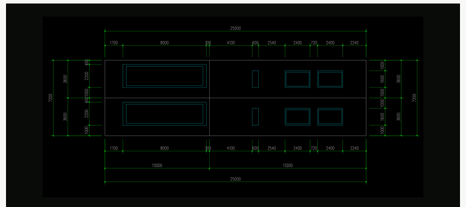
Complete drawing set



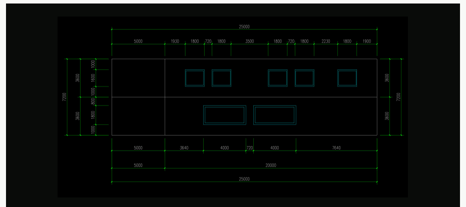
一层 / Floor 1



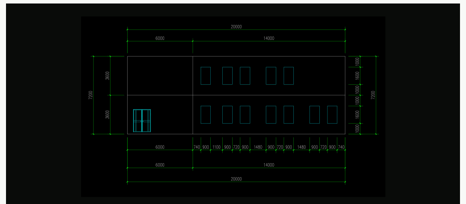
二层 / Floor 2



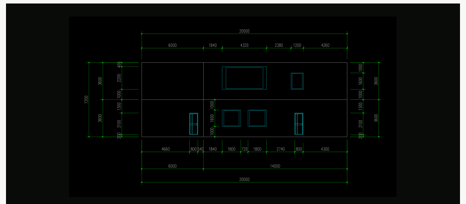
北 / North



南 / South



东 / East



西 / West

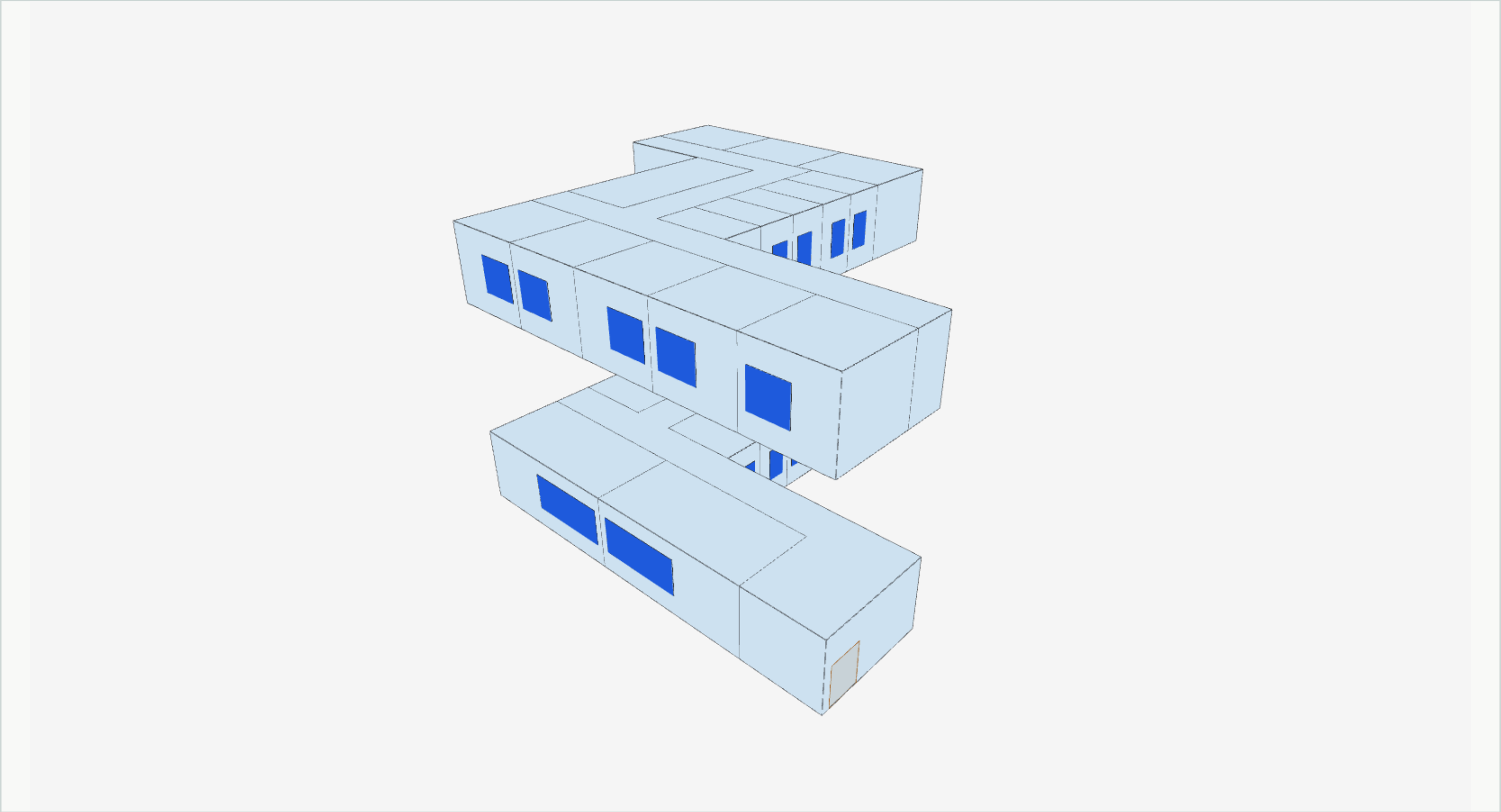
→

2 张平面 + 4 张立面

2 floor plans + 4 elevations

轻量 BIM

Lightweight BIM



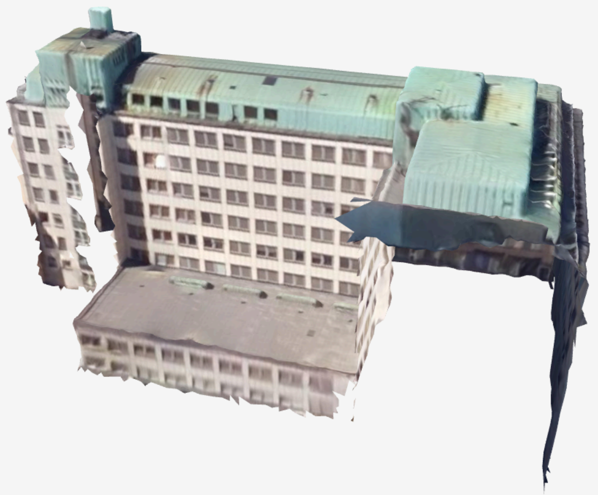
29 空间 Spaces　31 窗 Windows　30 门 Doors

真实贴图体量 → 轻量 BIM

A real textured mesh to lightweight BIM

真实贴图体量

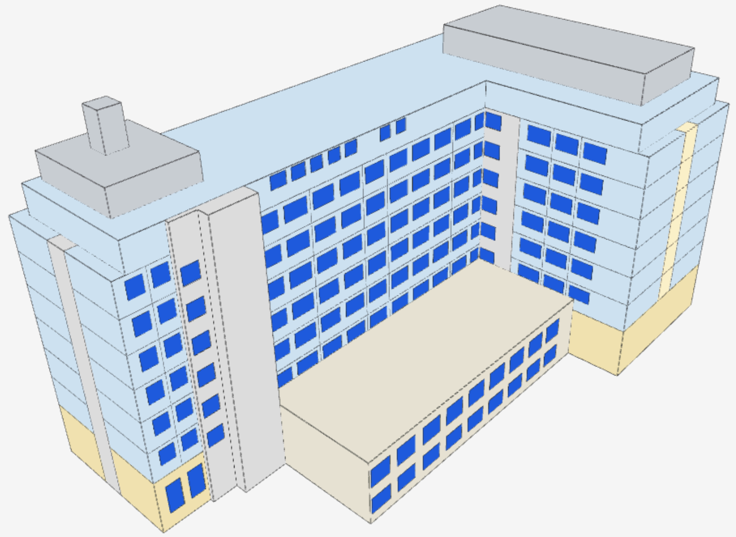
Textured mesh · GLB



→

轻量 BIM

Lightweight BIM



Voimatalo · Helsinki

摄影测量 / Photogrammetry · SUM / Helsinki

8 主楼层数 Main storeys

322 窗组 Window groups

01

提升空间理解能力

Stronger spatial understanding

空间 · 连接 · 建筑逻辑

Spaces · Connectivity
Architectural logic

02

提高复杂建筑稳定性

Robustness on complex buildings

跨案例 · 跨形态

Across building cases
and architectural forms

03

降低模型依赖度

Less reliance on large models

工具与轻量模型协同

Geometry tools
& smaller models

04

轻量 BIM 交互式编辑

Interactive lightweight BIM editing

查看 · 修改 · 反馈

Inspect · Edit
Feed back into design